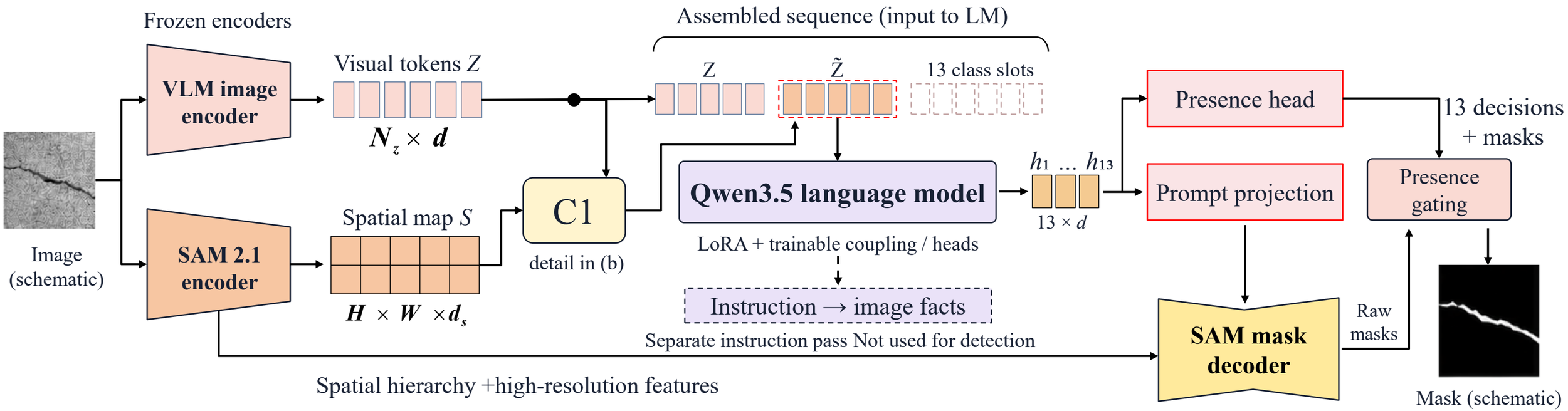
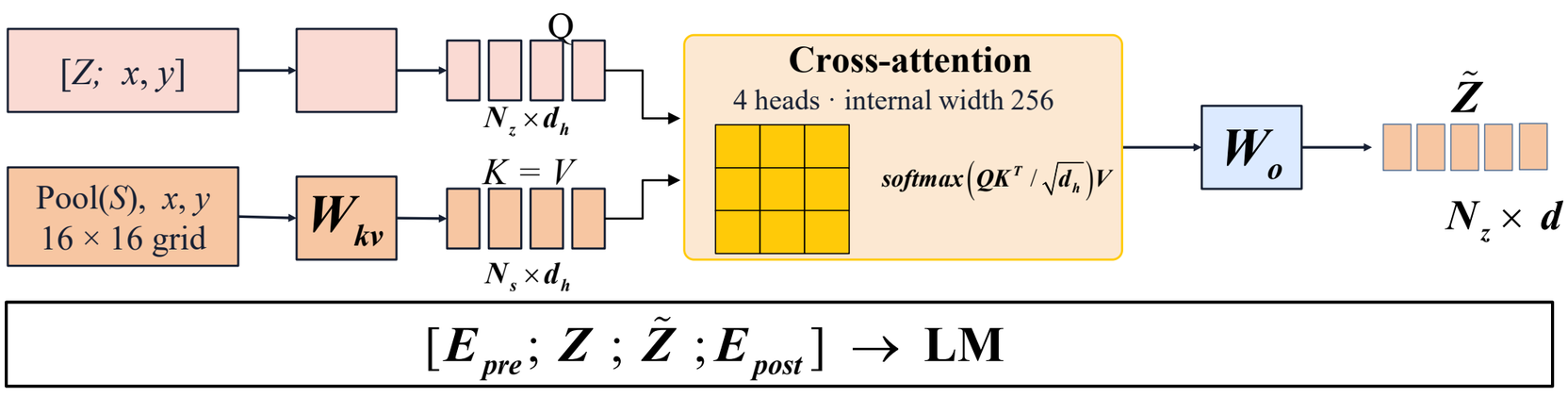


(a) TunnelWeave overview

Perception uses only the image and fixed class slots.



Shared normalized coordinates. Original Z retained.

(b) C1: coordinate alignment and token append

Same images, seeds, 1,600 updates

Model	Change relative to M
M: full	Semantic h Text CE on
Q: visual path	15 image-conditioned queries replace h. CE on
I: instruction	Same visual path and image supervision. CE off

(c) Matched controls